Constructing Solar Panel Waste Recycling Facilities in Massachusetts

Gozel Dovranova, Rinoa Oliver, Nilay Mishra

15.C571 Final Project

1 Problem Description

With a global transition to renewable energy, the number of photovoltaic (PV) solar panels is rising exponentially. However, this success creates a significant operational challenge: the management of End-of-Life (EOL) PV panels. With an average functional lifespan of approximately 15 years, the first generation of solar panels is now entering waste streams [2]. As a result, current projections estimate that global PV panel waste could reach 78 million tonnes by 2050 [3]. Despite challenges associated with EOL management, these PV modules present significant economic opportunities. They contain valuable elements such as silver, silicon, and copper. Effective recovery of these materials offers the potential to transform currently hazardous waste into profitable assets.

The recycling of these panels requires the construction of specialized recycling facilities. However, solar panels are dispersed across thousands of residential and commercial installations. Consequently, collection and transportation costs can account for up to 60% of the total recycling cost, which can even exceed the profit from recovered materials [7]. This challenge creates an opportunity for the application of optimization methods. In this study, we address this issue through a Facility Location Problem (FLP) within the context of Massachusetts. By optimizing recycling facility locations to minimize transportation costs, this work aims to lower barriers to recycling and improve both the economic and environmental benefits of solar panels.

2 Methods: Pre-Processing / Google API

Some pre-processing was required to get proper data for the facilities and the existing plants. First, latitude and longitude data of the existing plants was matched with a polygonal representation of Massachusetts state boundaries, to determine which plants were in Massachusetts. Also, since our data was in multiple overlapping sources, we had to merge the datasets together.

To determine possible existing facilities, we took a bounding box of the Massachusetts state

lines and sampled points inside it in a grid-like manner (so they would be roughly equally spaced). However, because of the shape of the Earth, a "latitude correction factor" of $\cos\left(\frac{\theta_{\min}+\theta_{\max}}{2}\right)$ was needed. This was because while a degree of latitude represents roughly the same amount of distance, a degree of longitude varies in distance depending on the current value of the latitude. Once these grid points were determined, the Massachusetts boundary polygon was used again to filter for the points within the state border. The resulting process gave us the coordinates for 95 possible facilities, roughly spaced equally across the state.

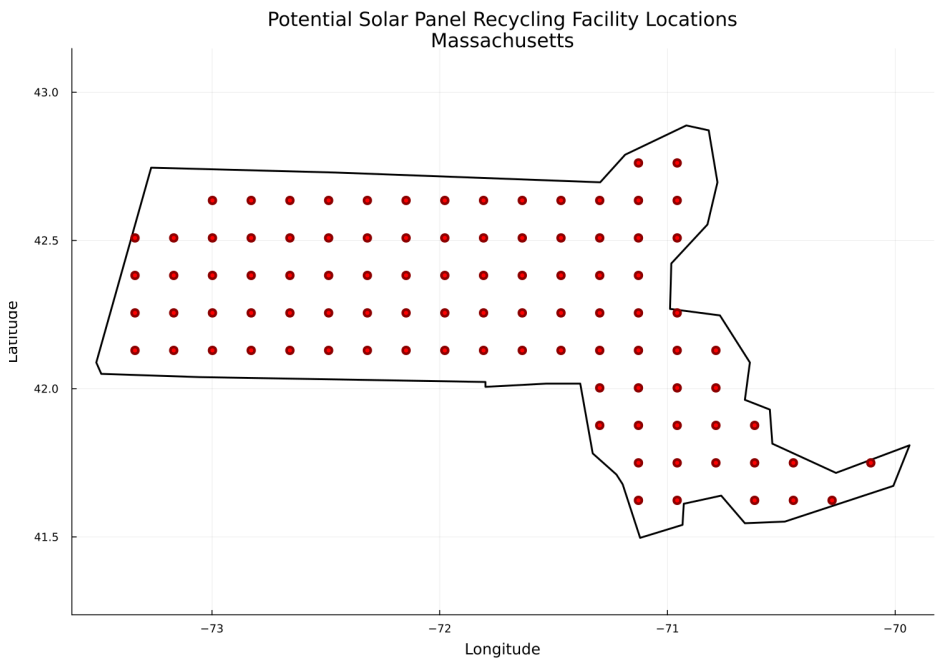


Figure 1: Map of Potential Waste Facility Locations

We also had to determine the distances of the form $\text{dist}_{i,s}$ introduced in Section 4.1. In order to do this, we used the Distance Matrix API provided by Google Maps. Between every pair of the 95 possible facility locations that we generated and the 338 existing plant locations, we sent an API query to Google Maps to determine the total transport distance. To speed up the ≈ 32000 API calls, we used parallel computation methods to split the work (since each call was independent).

3 Methods: Cost Estimates

The construction cost of a recycling facility with an annual processing capacity between 1,000 and 5,000 tons is estimated to range from \$200,000 to \$700,000 [5, 4]. This gives us cost of \$125 per additional ton capacity. The processing cost is approximately \$500 per ton, excluding transportation expenses [6]. The estimated value of recovered materials is \$800 per ton. Transportation costs are estimated at \$3.14 per mile [1].

4 Methods: Optimization Algorithm

The math governing our optimization algorithm is described in this section.

4.1 Parameters

- F_c: Fixed cost to open a facility: \$200,000
- V_c: Variable operating cost per additional ton of waste: \$125
- T_c: Cost per ton-mile of transporting waste: \$3.14
- $\delta = 0.95$: Annual discount factor for facility fixed and variable costs.
- $\text{tons_waste}_s \geq 0 \ \forall s$: Waste in tons generated by plant s, computed as plant capacity in MW times 200 tons per MW [3] [8]
- $\text{year_exps}_s \in \{1, \ldots, 10\} \ \forall s$: Expiration year index for plant s, computed by year plant opens minus 2009, as plants open between 2010 and 2019.
- $\text{dist}_{i,s} \geq 0 \ \forall i, s$: Distance in miles from facility i to plant s using Google API data.

4.2 Indices and Sets

- $i \in \{1, \ldots, 95\}$: Candidate facility locations
- $s \in \{1, \ldots, 338\}$: Solar plant locations
- $t \in \{1, \ldots, 10\}$: Years (2010 to 2019, where $t = 1$ is 2010).

4.3 Decision Variables

- $x_{i,t} \in \{0, 1\} \ \forall i, t$: 1 if facility i is operating in year t, 0 otherwise.
- $y_i \geq 0 \ \forall i$: Capacity (in tons/year) of facility i.
- $z_{s,i,t} \geq 0 \ \forall s, i, t$: Tons of waste transported from plant s to facility i in year t.
- $e_i \in \{0, 1\} \ \forall i$: 1 if facility i is ever opened in any year, 0 otherwise.
- $b_{i,t} \in \{0, 1\} \ \forall i, t$: 1 if facility i has been opened by or in year t (cumulative indicator), 0 otherwise.
- $a_{i,t} \in \{0, 1\} \ \forall i, t$: 1 if facility i first opens in year t, 0 otherwise.
- $w_{i,t} \geq 0 \ \forall i, t$: Linearizing variable for $x_{i,t} \cdot y_i$ (operating capacity in year t).

4.4 Objective Function

We minimize the total cost over all facilities of (1) the discounted fixed cost incurred when a facility first opens, plus (2) the discounted variable operating costs per year based on operating capacity, plus (3) transportation costs for shipping waste from all plants to the facility (assumed to be constant regardless of year):

$$\min \sum_{i=1}^{95} \left[\sum_{t=1}^{10} a_{i,t} \cdot \delta^t \cdot F_c + \sum_{t=1}^{10} w_{i,t} \cdot \delta^t \cdot V_C + \sum_{t=1}^{10} \sum_{s=1}^{338} z_{s,i,t} \cdot dist_{i,s} \cdot T_c \right]$$

4.5 Constraints

4.5.1 Facility Existence

We must ensure that if a facility operates in any year, ($x_{i,t} = 1$), then it must exist ($e_i = 1$).

$$e_i \geq x_{i,t} \quad \forall i = 1, \ldots, 95, \; t = 1, \ldots, 10$$

Additionally, we must ensure that a facility exists ($e_i = 1$) if and only if it operates in at least one year:

$$e_i \leq \sum_{t=1}^{10} x_{i,t} \quad \forall i = 1, \ldots, 95$$

4.5.2 Capacity Bounds

We must limit the capacity of a facility to at most big M (taken as the sum of total waste in all years, plus 1) if it exists ($e_i = 1$), and otherwise the capacity must be 0:

$$y_i \leq M \cdot e_i \quad \forall i = 1, \ldots, 95$$

We also enforce a minimum capacity of 1 if the facility exists:

$$y_i \geq e_i \quad \forall i = 1, \ldots, 95$$

4.5.3 Indicator on whether Facility has Opened yet

We set the indicator on whether the facility has opened yet in year 1 equal to whether the facility operates in year 1:

$$b_{i,1} = x_{i,1} \quad \forall i = 1, \ldots, 95$$

For years after 1, we ensure that the opening indicator cannot decrease, so once it becomes 1, it remains 1:

$$b_{i,t} \geq b_{i,t-1} \quad \forall i = 1, \ldots, 95, \; t = 2, \ldots, 10$$

For years $t = 2, ..., 10$, we require the opening indicator in year t to be 1 if the facility operates in year t:

$$b_{i,t} \geq x_{i,t} \quad \forall i = 1, \ldots, 95, \; t = 2, \ldots, 10$$

We need to prevent the indicator from becoming 1 unless it was already equal to 1 in the previous year, or the facility is operating in the current year:

$$b_{i,t} \leq b_{i,t-1} + x_{i,t} \quad \forall i = 1, \ldots, 95, \ t = 2, \ldots, 10$$

4.5.4 First Opening Year Indicator

We ensure that this indicator is 1 in exactly one year if the facility exists ($e_i = 1$), otherwise, all $a_{i,t} = 0$.

$$\sum_{t=1}^{10} a_{i,t} = e_i \quad \forall i = 1, \ldots, 95$$

We ensure that the facility cannot first open in year t unless it is open in t. If a facility opens, $a_{i,t} = 1$ in only that opening year, and $b_{i,t} = 1$ for that year and all following years, so thus:

$$a_{i,t} \leq b_{i,t} \quad \forall i = 1, \ldots, 95, \ t = 1, \ldots, 10$$

If a facility opens in year 1, both variables will be equal to 1, and otherwise they will both be 0, so we ensure that:

$$a_{i,1} = b_{i,1} \quad \forall i = 1, \ldots, 95$$

f the cumulative indicator switches from 0 to 1 in year t, then we should have $a_{i,t} = b_{i,t} - b_{i,t-1} = 1$, and otherwise, we should have that $a_{i,t} = b_{i,t} - b_{i,t-1} = 0$. Thus:

$$a_{i,t} = b_{i,t} - b_{i,t-1} \quad \forall i = 1, \ldots, 95, \ t = 2, \ldots, 10$$

4.5.5 Linearization of Capacity times Yearly Activity Variable

We must ensure that capacity is less than or equal to the big M if the facility is active in that year, and otherwise capacity times binary activity variable in that year is 0:

$$w_{i,t} \leq M \cdot x_{i,t} \quad \forall i = 1, \ldots, 95, \ t = 1, \ldots, 10$$

We need to ensure capacity doesn't exceed facility capacity:

$$w_{i,t} \leq y_i \quad \forall i = 1, \ldots, 95, \ t = 1, \ldots, 10$$

We force $w_{i,t} = yi$ when $x_{i,t} = 1$:

$$w_{i,t} \geq y_i - M \cdot (1 - x_{i,t}) \quad \forall i = 1, \ldots, 95, \ t = 1, \ldots, 10$$

4.5.6 Waste Allocation

We prevent waste from plant s to be transported to any waste facility in years other than its expiration year:

$$\sum_{j=1}^{95} z_{s,j,t} = 0 \quad \forall s = 1, \ldots, 338,\ t = 1, \ldots, 10 \text{ where } t \neq \text{year_exps}_s$$

We ensure that all waste from plant s is fully sent to waste facilities in its expiration year:

$$\sum_{j=1}^{95} z_{s,j,t} = \text{tons_waste}_s \quad \forall s = 1, \ldots, 338,\ t = 1, \ldots, 10 \text{ where } t = \text{year_exps}_s$$

We limit waste from plant s to facility j in its expiration year to facility j's capacity:

$$z_{s,j,t} \leq y_j \quad \forall s = 1, \ldots, 338,\ j = 1, \ldots, 95,\ t = 1, \ldots, 10 \text{ where } t = \text{year_exps}_s$$

4.5.7 Waste Processing Limited by Operating Capacity

We ensure that the total waste processed at facility i in year t can't exceed its operating capacity in that year:

$$\sum_{\substack{s=1 \\ \text{year_exps}_s = t}}^{338} z_{s,i,t} \leq w_{i,t} \quad \forall i = 1, \ldots, 95,\ t = 1, \ldots, 10$$

5 Results

The total cost of constructing the facilities, operating them, and shipping waste to them is an estimated **$30.9 million dollars**. The majority of this cost, around 20.2 million (65%) comes from maintaining the facility, while 4.2 million (14%) comes from the fixed construction costs, and 6.5 million (21%) from transportation costs.[1]

Across all 10 years from when the first plants in Massachusetts expire (2025-2034), **25 total facilities** were constructed, with an average capacity size of 2,352 tons:

[1]Note that transportation costs are roughly 40%-60% of processing costs for waste. Therefore, if we included processing costs, we would likely incur 36.1-38.7 million dollars in total costs.

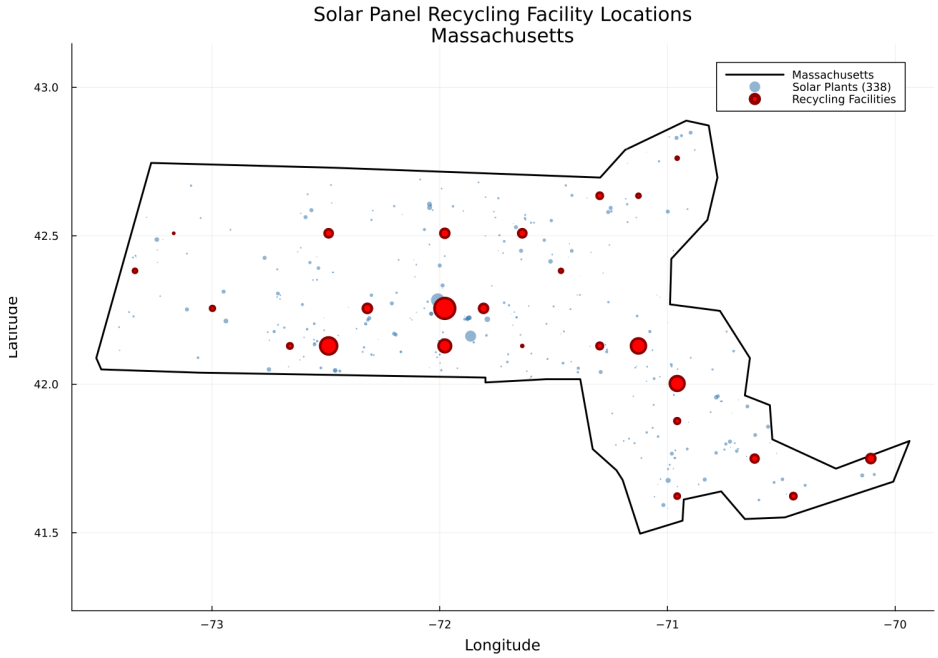
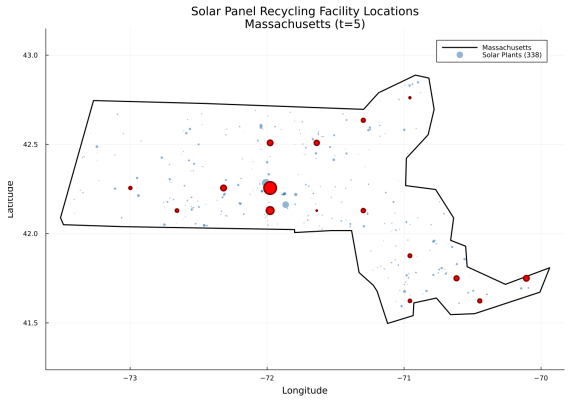
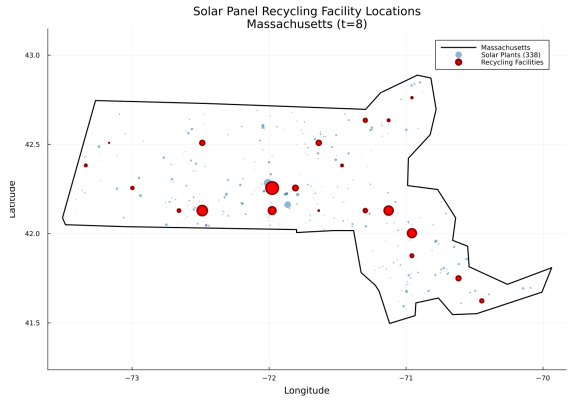


Figure 2: All 25 Recycling Facilities Created Across Decade

All recycling facilities were used in multiple years. Most recycling facilities had years of inactivity in between years in which waste was actively sent to them. For instance, 15 recycling facilities were used in year 2029 (left) and 21 were used in year 2032 (right):

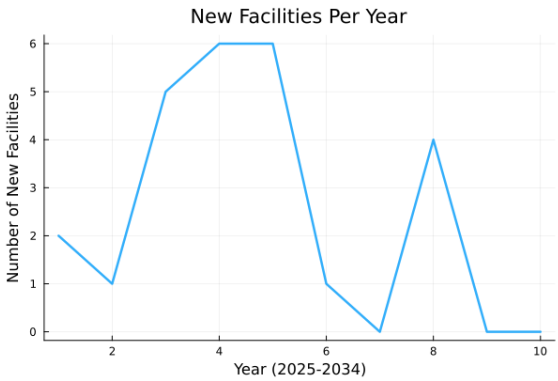


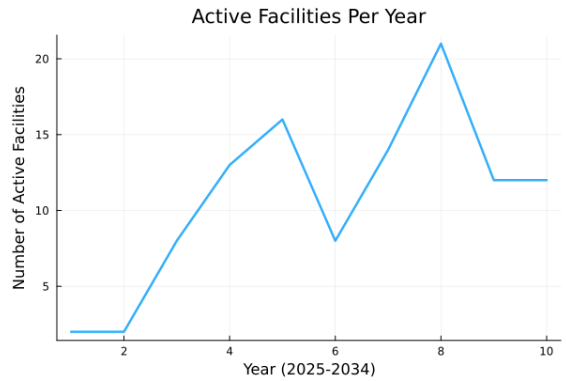
Facilities used in 2029



Facilities used in 2032

The creation of new facilities is centered around the earlier half of the decade, with no new facilities created in 2033 or 2034. The period in which the greatest number of facilities are actively collecting waste is near the end of the decade, peaking in 2032:





These results indicate how we may prioritize constructing certain facilities over others depending on how critically they are needed in the near future. For instance, the model only constructed two facilities in 2025 and one additional facility in 2026. Additionally, they showcase the benefits of creating facilities and temporarily leaving them inactive depending on yearly waste flow before reusing them again.

6 Discussion

As previously stated, recycling waste facilities would be a novel technology which can provide substantial environmental benefits in handling the waste from expired solar plants. This model may aid Massachusetts policy makers in determining where to build recycling facilities, which facilities should be prioritized in construction based on the years in which our model predicts they are needed, and the size of these facilities. Additionally, we may provide Massachusetts policy makers with rough estimates on the budget needed to build these facilities.

This project could be developed further by running our algorithm on the data set of solar panels for the entirety of the United States (over three thousand solar panels) as opposed to just Massachusetts. Unfortunately, running both the Google Maps API algorithm and the optimization algorithm would likely be incredibly time-consuming.

Additionally, we may include strict capacity constraints on facility sizing. As opposed to a fixed cost of $\$200,000$ to open any facility, we can assign this fixed cost to the smallest feasible waste facility, estimated to process up to around 1,000 tons. Our upper bound on sizing was estimated to be around 5,000 tons; fortunately, the waste facilities our model constructed process is consistently less than this upper bound.

For simplicity, as it would be difficult to obtain precise land costs in every one of the 95 potential waste locations, we assumed that the facility cost with respect to size would be equivalent in every location. Varying costs provide an additional means to make our model more realistic. In particular, it would create a tradeoff between building several cheap facilities in suboptimal locations versus a single, more expensive one in a better location. It would also help to make sure our waste locations are also "valid" locations to build a plant e.g. the land is public or for sale.

As our model is directed towards policy makers, we haven't included analysis of potential profit from recycled materials if they are integrated into the economy. Our model may benefit

from including these estimates, in addition to the budget needed to construct and operate these facilities.

References

[1] Appropedia contributors, *"PV recycling: FAST literature reviews,"*, Appropedia, 2025.

[2] International Renewable Energy Agency (IRENA), *End-of-Life Management: Solar Photovoltaic Panels*, IRENA, Abu Dhabi, 2016.

[3] H. Weckend, A. Wade, and G. Heath, *End-of-Life Management of Solar Photovoltaic Panels*, International Renewable Energy Agency, 2016.

[4] Okon Recycling, *Solar Panel Recycling Cost*, Okon Recycling, 2023.

[5] Solutions for E-Waste, *Solar Panel Recycling Plant Cost*, Solutions for E-Waste, 2023.

[6] UNSW Sydney, Australian Centre of Advanced Photovoltaics (ACAP), *"Bigger and better solar panel recycling centres needed to deal with PV waste,"*, UNSW Newsroom, 2024.

[7] Y. Xu, B. Li, and L. Zhang, *Economic Analysis of Solar Panel Recycling and Transportation Costs*, Renewable Energy, vol. 146, pp. 2350–2359, 2020.

[8] J.M. Mason, V.M. Fthenakis, T. Hansen, and H.C. Kim, *Energy Pay-Back and Life Cycle CO2 Emissions of the BOS in an Optimized 3.5 MW PV Installation*, Progress in Photovoltaics Research and Applications, 14:179-190, 2006.